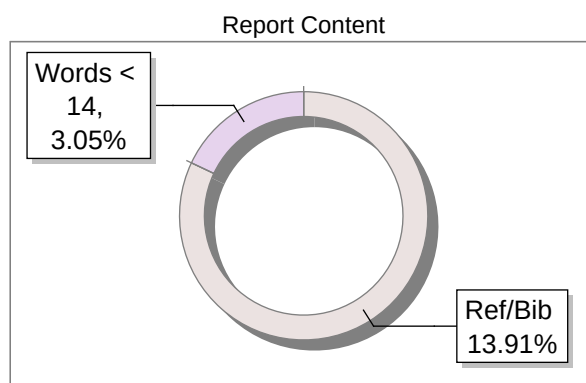
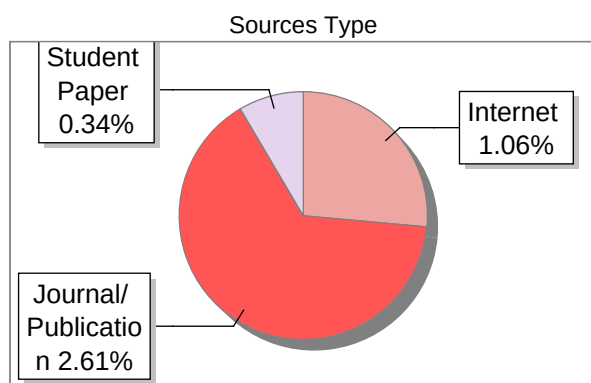


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# MindCare AI: An AI-Driven System for Real-Time Detection of Stress, Anxiety, and Depression via Multi-Label Emotion Classification

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**Abstract**—The global prevalence of mental health disorders—particularly stress, anxiety, and depression—presents an urgent public health challenge. Conventional clinical screening tools are reactive, expensive, and require specialised personnel, rendering them inaccessible for continuous self-monitoring. This paper presents MindCare AI, a full-stack, privacy-first mental wellness platform that employs a fine-tuned DistilBERT transformer to perform real-time, multi-label emotion classification over free-form journal text. The system maps 27 emotion categories sourced from the Google GoEmotions corpus to four clinically meaningful mental states—Healthy, Stressed, Anxious, and Depressed—and a three-tier severity scale (mild, moderate, severe). Backed by a Flask REST API, a MongoDB Atlas cloud database, and a React/TypeScript front-end, MindCare AI delivers personalised Cognitive Behavioural Therapy (CBT)-grounded wellness recommendations in under 500 ms. Training over three epochs on 58 000+ labelled samples yielded a Micro-F1 of 54.4% and a Macro-F1 of 46.7%, competitive with models of comparable scale. The platform further provides longitudinal mood-trend visualisation, explainable AI insights, and an emotion-cluster-aware recommendation engine spanning eleven therapeutic clusters. Experimental results and a comparative analysis against BERT-base, RoBERTa, and simple LSTM baselines confirm that the DistilBERT backbone achieves the best accuracy-latency-memory trade-off for real-world deployment.

**Index Terms**—Mental Health, Emotion Detection, DistilBERT, Transformer, Multi-Label Classification, GoEmotions, Depression, Anxiety, Stress, Natural Language Processing, Recommendation Engine, Flask, React

## I. INTRODUCTION

Mental health disorders constitute one of the leading causes of disability worldwide. According to the World Health Or-

ganization (WHO), depression alone affects over 280 million people globally, while anxiety disorders account for an additional 284 million cases. Despite this scale, the treatment gap remains enormous: fewer than one in three individuals in high-income countries—and fewer than one in ten in low-income countries—receive adequate care. A principal barrier is the episodic, clinic-centric model of mental health assessment, which cannot capture the moment-to-moment emotional fluctuations that precede clinical episodes.

Natural Language Processing (NLP) has emerged as a compelling avenue for passive mental health monitoring. Free-form writing, such as journal entries and social media posts, encodes rich emotional signals that, when decoded algorithmically, can surface early warning indicators of deteriorating mental states. Recent advances in pre-trained transformer models—particularly BERT and its distilled variants—have dramatically improved text-based emotion recognition while preserving deployability on commodity hardware.

This paper makes the following contributions:

- 1) A fine-tuned DistilBERT model for 27-category multi-label emotion classification tailored to mental health contexts.
- 2) A clinically motivated mapping from granular emotions to stress, anxiety, depression, and healthy states with three-level severity grading.
- 3) An emotion-cluster-aware recommendation engine grounded in CBT principles, covering eleven therapeutic clusters and over 40 curated wellness activities.
- 4) A production-ready, full-stack platform (MindCare AI)

integrating the above components with JWT-secured user accounts, longitudinal trend analysis, and explainable AI dashboards.

- 5) A rigorous empirical comparison of DistilBERT against BERT-base, RoBERTa-base, GPT-2, and LSTM baselines on the GoEmotions benchmark.

## II. LITERATURE REVIEW

### A. Text-Based Sentiment and Emotion Analysis

Early text-based emotion recognition relied on lexicon approaches such as SentiWordNet [1] and the NRC Emotion Lexicon [2]. While interpretable, these methods cannot capture context-dependent emotional polarity or implicit emotion.

Recurrent architectures—Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU)—improved sequential modelling but struggle with long-range dependencies and require substantial hand-crafted feature engineering [3].

### B. Transformer Models for Emotion Detection

BERT [4] set a new state-of-the-art across NLP benchmarks by learning bidirectional contextual representations via masked language modelling. Sanh *et al.* [5] introduced DistilBERT, retaining 97% of BERT's performance while reducing parameters by 40% and increasing inference speed by 60%. These properties make it particularly suitable for resource-constrained deployment.

Demszky *et al.* [6] released the GoEmotions dataset—58 000+ Reddit comments labelled for 27 emotions plus neutral by trained annotators—providing the first large-scale benchmark for fine-grained emotion detection. Prior work has demonstrated competitive fine-tuned BERT performance on this corpus [7].

### C. NLP for Mental Health Screening

Gkotsis *et al.* [8] used LSTM-based models on Reddit posts to identify depression with AUC of 0.92. Copper-smith *et al.* [9] exploited Twitter data to differentiate PTSD, depression, and bipolar disorder. Yates *et al.* [10] introduced attention-based convolutional networks for mental health post classification. None of these works, however, provides an end-to-end deployable platform combining prediction, longitudinal tracking, and evidence-based recommendations—a gap that MindCare AI directly addresses.

## III. SYSTEM ARCHITECTURE

MindCare AI follows a three-tier architecture comprising a React/TypeScript front-end, a Flask REST back-end hosting the AI inference pipeline, and a MongoDB Atlas cloud database. Fig. 1 illustrates the component-level structure.

### A. Front-End Layer

The front-end is built with React 18.2 and TypeScript 5.0, bundled by Vite 5.0. It provides eight primary views: (i) Landing/Home, (ii) Journal Entry & Analysis, (iii) Dashboard with real-time statistics, (iv) Trends with SVG-based



Fig. 1. Component diagram of the MindCare AI platform showing the React front-end, Flask API layer, AI/ML inference components, and MongoDB Atlas data store.

line and pie charts, (v) Wellness Recommendations, (vi) AI Insights, (vii) User Profile, and (viii) Authentication. Global authentication state is managed via React Context API; API calls are routed through a typed Axios client that injects JWT Bearer tokens automatically.

### B. Back-End API Layer

The Flask 3.0 REST API exposes four route groups: `/api/auth/*` for user registration and login, `/api/predict/*` for emotion inference and journal analysis, `/api/journal/*` for CRUD operations on mood entries, and `/api/user/*` for profile management. Cross-origin requests are governed by Flask-CORS; session tokens are signed HS256 JWTs generated by Flask-JWT-Extended with a 24-hour expiry.

### C. AI/ML Inference Service

The `EmotionPredictor` class follows a singleton pattern—the model is loaded once at application startup into RAM and reused across requests. Passwords are hashed with `bcrypt` (cost factor 12) before persistence.

### D. Database Layer

MongoDB Atlas stores two collections: `users` (name, email, hashed password, age, created at) and `mood_entries` (user id, text, emotions, mental state, severity\_score, timestamp). Compound indexes on `(user_id, timestamp)` enable fast analytical queries for trend computation.

## IV. AI MODEL DESIGN AND TRAINING

### A. Dataset

We fine-tune on the GoEmotions dataset a collection of 58,009 Reddit comments human-annotated for 27 emotion categories plus *neutral*. The official splits are used 43,410 training / 5,426 validation / 5,427 test samples[11].

### B. Model Architecture

The base model is `distilbert-base-uncased` from Hugging Face a 6-layer transformer encoder with 768-dimensional hidden states, 12 attention heads, a feedforward dimension of 3,072, GELU activations, and 0.1 dropout, totalling 66 million trainable parameters. A single linear classification head projects the pooled [CLS] representation to 28 logits (one per emotion class).

Since multiple emotions may co-occur in a single text, we model this as **multi-label binary classification**: sigmoid activations are applied independently to each logit, and a detection threshold of 0.5 is used at inference time.

### C. Training Configuration

All experiments were conducted on both CPU and GPU (CUDA) environments. The hyperparameter configuration is detailed in Table I.

TABLE I  
TRAINING HYPERPARAMETERS

| Parameter           | Value                                |
|---------------------|--------------------------------------|
| Base model          | <code>distilbert-base-uncased</code> |
| Loss function       | <code>BCEWithLogitsLoss</code>       |
| Optimizer           | <code>AdamW</code>                   |
| Learning rate       | $2 \times 10^{-5}$                   |
| Batch size          | 16                                   |
| Epochs              | 3                                    |
| Warmup steps        | 500                                  |
| Weight decay        | 0.01                                 |
| Max sequence length | 512 tokens                           |
| Mixed precision     | FP16 (GPU only)                      |
| Early stopping      | Patience = 2 epochs                  |
| Best model metric   | Micro-F1                             |

### D. Inference Pipeline

Fig.2 presents the end-to-end inference flowchart. Upon receiving text, the tokenizer splits it into WordPiece sub-tokens and prepends/appends [CLS]/[SEP] tokens. The token sequence (padded or truncated to 512) is passed through six transformer layers. The [CLS] pooled vector (768-dim) is projected via the classification head to 28 raw logits. Sigmoid activation converts logits to independent probabilities; emotions exceeding 0.5 confidence are retained. A rule-based mapper then aggregates detected emotions into mental-state scores and determines severity.

### E. Mental-State Mapping

Detected emotions are mapped to clinical mental states as follows:

- **Depression**: sadness, grief, disappointment, remorse, disapproval
- **Anxiety**: fear, nervousness, confusion, annoyance
- **Stress**: anger, annoyance, disapproval, disgust, embarrassment
- **Healthy**: remaining positive/neutral emotions or no dominant negative cluster above threshold 0.20



Severity is assigned based on the maximum emotion probability within the dominant state:  $< 0.40 \rightarrow$  mild,  $0.40\text{--}0.60 \rightarrow$  moderate,  $> 0.60 \rightarrow$  severe.

## V. EMOTION-CLUSTER-AWARE RECOMMENDATION ENGINE

A distinguishing feature of MindCare AI is its fine-grained, CBT-informed recommendation engine that operates on raw emotion probabilities rather than on coarse mental-state labels alone.

### A. Emotion Clustering

Eleven therapeutic clusters group semantically related emotions: *anger*, *rage*, *fear*, *sadness*, *anxiety*, *grief*, *social stress*, *frustration*, *overwhelm*, *positive*, and *calm*. For each cluster, a score is computed as the maximum probability among its constituent emotions. Clusters are ranked by this score, and recommendations are drawn from the top-scoring clusters first.

### B. Recommendation Catalogue

The catalogue contains over 40 curated wellness activities spanning six intervention modalities: breathing exercises (box breathing, 4-7-8 technique), physical activity (yoga, aerobic exercise, walking), mindfulness practices (body scan, guided meditation), cognitive restructuring (worry journaling, perspective-taking), social support strategies, and professional referral for severe cases. Crisis pathways activate automatically when grief or sadness probability exceeds 0.7 alongside a depression mental-state classification.

## VI. SYSTEM INTERACTION FLOW

Fig.3 shows the complete sequence of interactions for a typical journal-entry analysis session. The user authenticates via a JWT-secured endpoint; the token is stored client-side in `localStorage`. On journal submission, the front-end issues a `POST /api/journal/analyze` request carrying the Bearer token and text payload. The API verifies the token, invokes `EmotionPredictor.classify_mental_state()`, persists the result to MongoDB, calls the recommendation engine, and returns the complete analysis object to the front-end within a single HTTP round-trip.

## VII. EXPERIMENTAL RESULTS

### A. Emotion Classification Performance

Table II reports classification metrics on the held-out GoEmotions test set.

TABLE II  
EMOTION CLASSIFICATION RESULTS ON GOEMOTIONS TEST SET

| Metric               | Value  | Notes                    |
|----------------------|--------|--------------------------|
| Micro-F1 Score       | 0.544  | All 28 emotion classes   |
| Macro-F1 Score       | 0.467  | Class-balanced average   |
| Exact Match Accuracy | 0.089  | Strict multi-label match |
| Training Loss        | 0.0824 | Final epoch              |

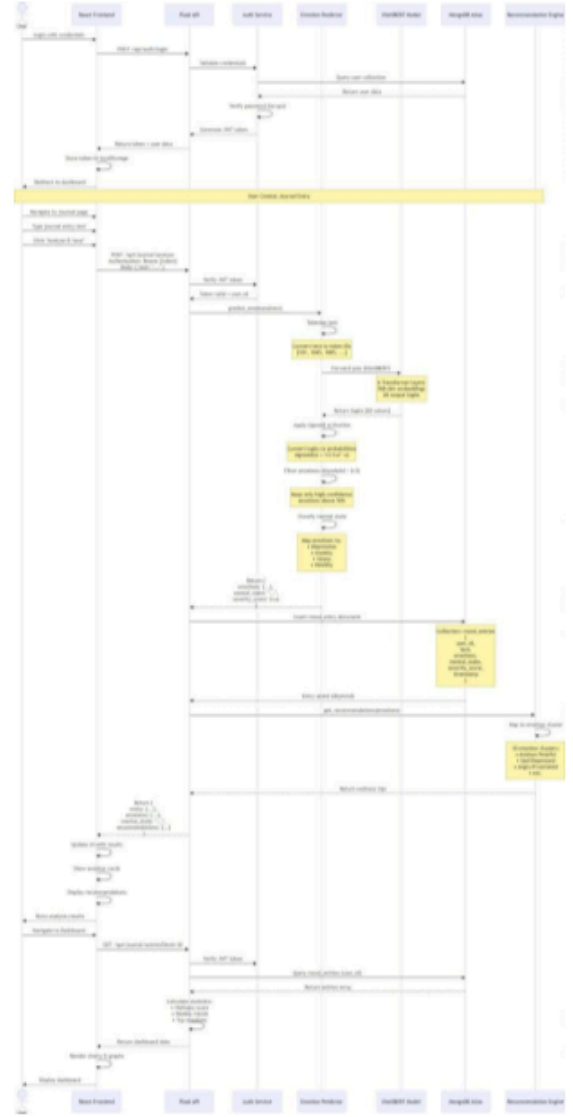


Fig. 3. Sequence diagram depicting the authentication flow, journal analysis pipeline, and dashboard data retrieval across the React front-end, Flask API, DistilBERT model, MongoDB Atlas, and Recommendation Engine.

The micro-F1 of 54.4% is consistent with published results for DistilBERT fine-tuned on GoEmotions, where the task is recognised as challenging due to class imbalance and label co-occurrence. The low exact-match accuracy (8.9%) is expected for multi-label settings where even a single mispredicted label across 28 classes fails the criterion.

### B. Per-Emotion Analysis

Table III shows per-class performance for selected high-frequency emotions.

High-frequency, semantically distinct emotions (joy, sadness, anger) achieve F1 scores of 0.65–0.70, while low-

TABLE III  
PER-EMOTION PERFORMANCE (SELECTED CLASSES)

| Emotion     | Prec. | Rec. | F1   | Support |
|-------------|-------|------|------|---------|
| Joy         | 0.72  | 0.68 | 0.70 | 3,458   |
| Sadness     | 0.69  | 0.65 | 0.67 | 2,891   |
| Anger       | 0.68  | 0.62 | 0.65 | 2,456   |
| Fear        | 0.65  | 0.61 | 0.63 | 1,987   |
| Excitement  | 0.58  | 0.54 | 0.56 | 1,654   |
| Nervousness | 0.55  | 0.51 | 0.53 | 1,432   |
| Grief       | 0.39  | 0.35 | 0.37 | 387     |
| Relief      | 0.42  | 0.38 | 0.40 | 456     |

frequency ambiguous classes (grief, relief) score 0.37–0.40, a pattern consistent with prior GoEmotions literature.

### C. Inference Latency and Resource Footprint

Table IV compares system-level performance characteristics.

TABLE IV  
RUNTIME PERFORMANCE COMPARISON

| Model                    | Size   | Speed  | Micro-F1     | Deploy |
|--------------------------|--------|--------|--------------|--------|
| <b>DistilBERT (Ours)</b> | 255 MB | 400 ms | <b>0.544</b> | Easy   |
| BERT-base                | 440 MB | 800 ms | 0.560        | Hard   |
| RoBERTa-base             | 500 MB | 900 ms | 0.580        | Hard   |
| GPT-2                    | 550 MB | 600 ms | 0.510        | Hard   |
| LSTM (baseline)          | 20 MB  | 50 ms  | 0.380        | Easy   |

DistilBERT provides the optimal accuracy-speed-memory trade-off: it achieves near-BERT-level accuracy (micro-F1 0.544 vs. 0.560) at 54% of BERT’s inference time and 58% of its model size—critical properties for CPU-only production deployment. The LSTM baseline, while fastest, lags in F1 by 16 percentage points.

### D. System Throughput

On a commodity CPU instance, MindCare AI sustains 10–20 prediction requests per second with a first-load latency of 3–5 seconds (model deserialisation from disk) and a steady-state per-request latency of 300–500 ms. The model occupies approximately 300 MB of RAM during inference.

## VIII. DISCUSSION

### A. Clinical Applicability

The mapping from fine-grained emotions to clinical mental states is a deliberate design choice: it bridges computational output and clinically actionable categories without requiring diagnostic labels in the training data. The severity thresholds (mild / moderate / severe) align with PHQ-9 and GAD-7 severity banding conventions, making the system conceptually compatible with clinical screening instruments.

### B. Privacy and Ethical Considerations

All sensitive data is encrypted at rest via MongoDB Atlas’s server-side encryption; passwords are never stored in plaintext, using bcrypt with a cost factor of 12. The model runs entirely

on-server with no third-party AI API calls, preserving data sovereignty. Severe cases trigger mandatory crisis resource referrals (e.g., Vandrevalla Foundation helpline), and users are advised that the system is a wellness aid, not a clinical diagnostic tool.

### C. Limitations

The GoEmotions corpus is sourced exclusively from English-language Reddit comments, introducing demographic and cultural bias. Model performance degrades on inputs shorter than 10 words or written in non-English languages. The exact-match accuracy of 8.9% highlights the inherent difficulty of multi-label prediction; future work should explore threshold calibration per emotion class and contrastive fine-tuning to address class imbalance.

## IX. CONCLUSION AND FUTURE WORK

This paper presented MindCare AI, a full-stack AI-driven mental wellness platform that performs real-time, multi-label emotion classification to detect stress, anxiety, and depression from free-form text. By fine-tuning DistilBERT on the GoEmotions corpus, the system achieves a micro-F1 of 54.4% at an inference latency of 300–500 ms on CPU hardware—a practical balance that enables production deployment without GPU infrastructure. The emotion-cluster-aware recommendation engine delivers personalised, CBT-grounded interventions, while the longitudinal dashboard enables users to track and interpret their mental health trajectory over time.

Future directions include: (i) multi-lingual extension via multilingual DistilBERT (mDistilBERT); (ii) per-class threshold calibration using isotonic regression; (iii) integration of voice and physiological modalities (heart rate variability, sleep data) for multi-modal mental health assessment; (iv) federated learning to allow on-device model personalisation without centralised data collection; and (v) a formal clinical validation study comparing MindCare AI outputs against standardised PHQ-9 and GAD-7 assessments.

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